BHAVISHYA GUPTA

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PROFESSIONAL SUMMARY

Final-year B.Tech student at IIT Kanpur (Chemical Engineering + IME Minor) with industry experience in ML pipelines, statistical modelling, and data privacy compliance. Awarded a PPO by ICICI Bank; national competition winner with projects spanning multimodal classification, crowd dynamics, and educational analytics. Proficient in Python, SQL, PySpark, and deep learning frameworks; mentored 160 students across ML programs. Seeking full-time roles in Data Science, ML Engineering, Data Analytics, or Business Analytics.

EDUCATION

Indian Institute of Technology (IIT) Kanpur — B.Tech, Chemical Engineering | Minor: IME | CPI: 8.13/10.0 *2022 – 2026*

St. Mary's Convent School, Gajraula — CBSE XII: 96.4% (2021) | CBSE X: 96% (2019)

WORK EXPERIENCE

ICICI Bank, India — Data Compliance & Privacy Intern | Pre-Placement Offer (PPO) Awarded *May '25 – Jul '25*

- Designed and built end-to-end data pipelines mapping 5 consent/rights-management event types across 6+ product lines (iMobile, cards, loans, Aadhaar e-KYC, CIBIL); modelled complex entity relationships and granular data-flow touchpoints at enterprise scale.
- Engineered a remediation framework with modular opt-in components, event-driven consent-withdrawal logging, data-minimisation controls, and cookie-layer architecture; translated 10+ global regulatory case studies (Google, Meta, LinkedIn, TikTok) into DPDP-aligned engineering specifications, actionable mitigations, and a structured compliance roadmap.
- Delivered an executive-ready technical audit report adopted for internal review of consent practices across digital and branch channels; directly strengthened system auditability, DPDP readiness, and data transparency across all customer-facing products.

PROJECTS

Statistical Similarity Measure for Crowd Dynamics | Prof. Shankar Prawesh, IIT Kanpur GitHub *Apr '24 – Jul '24*

- Tracked pedestrian movements across video frames with 92% accuracy using YOLOv8n; improved multi-object tracking by 15% via Ensemble Kalman Smoothing; implemented Steering, Social Force, and Predictive Planning models to simulate realistic pedestrian dynamics, achieving smooth, natural collision-avoidance trajectories suited for urban crowd management.
- Validated simulation performance using statistical metrics (entropy, mean error, standard deviation) and visualized real-time trajectory data to generate actionable urban safety insights; optimized models for practical crowd-management deployment.

Multimodal Binary Classification with Feature Fusion | CS771 | Prof. Piyush Rai, IIT Kanpur GitHub *Aug '24 – Nov '24*

- Designed a low-data ML pipeline for multimodal binary prediction, tokenizing emoticons and vectorizing deep embeddings and digit-string features into standardized numeric representations; rigorously benchmarked LSTM, SVM, Random Forest, XGBoost, and CatBoost across progressively varied training splits (20–100%) to assess generalization under data-scarce conditions.
- Fused modalities via standardization, concatenation, and PCA; achieved 98.8% validation accuracy on the best single-modality model and 94.5% accuracy with fused SVM, demonstrating robust and generalizable multimodal learning at minimal data volumes.

Data-Driven Analysis of Educational Outcome Trends | DMS692 | Prof. Nivedita Bhaktha, IIT Kanpur GitHub *Feb '25 – Apr '25*

- Cleaned NDAP education datasets; conducted EDA and built visualizations to surface cross-state trends and disparities; compressed 7 performance metrics to 3 interpretable indicators via PCA and Factor Analysis (84% variance retained); identified significant temporal and regional shifts (MANOVA/ANOVA, $p < 0.001$) to inform policy interventions and resource allocation.

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, Scala, C++, C
ML & AI Frameworks: Scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers, OpenCV, PySpark, MLlib, GraphX, Optuna
Data Engineering & Cloud: Apache Spark, Hadoop, Kafka, Hive, Airflow, Snowflake, AWS, MLflow
Dev Tools & Infrastructure: FastAPI, Pydantic, Docker, Kubernetes, Git, Jupyter Notebook, Web Scraping
BI & Visualization: Power BI, Tableau, MS Excel, LaTeX

ACHIEVEMENTS, LEADERSHIP & ACTIVITIES

- Ranked 1st in Evonik Sustainability Challenge (national, INR 1.5 lakh); 2nd in Kanopy Techno Optimization Challenge (INR 10,000).
- Secured All India Rank 4428 in Joint Entrance Examination Advanced 2022 among a pool of 1.55 Lakh shortlisted candidates.
- Core Team Member, GDG IIT Kanpur (Apr '24 – Mar '25): onboarded 30+ industry partners, boosted club growth by 40%, drove 300+ attendees to inaugural events, grew student involvement by 50%, sponsorships by 70%, and scaled social media following 5×.
- Project Mentor – Introduction to Machine Learning (SimTech, Chemineers Society, 140 students): guided students from Python basics to algorithms like Random Forest, achieving $R^2 = 0.88$ and accuracy = 0.91 on a Wastewater Treatment Plant dataset.
- Project Mentor – Active Learning (GDG, 20 students): built and benchmarked a custom CNN against ResNet50 on CIFAR-10 using uncertainty metrics (entropy, margin sampling) and diversity metrics (cosine similarity, KL divergence); 72% vs. 84% accuracy.
- Mentored 7 IIT Kanpur juniors through the competitive internship recruitment season, enabling successful offer conversions.
- Growth Intern, FACT APP (1 month): executed marketing strategy, mentored JEE aspirants, and drove network-building initiatives.